Physical Chemistry CUME, September 29, 2017

Quantum Chemistry

A particle of mass 1.0 MeV/c^2 is free within a 1D box of width 1.0 Å. The particle is not allowed to go outside the box. The energy of the particle was measured to be about 0.3 keV at time $t = 0$.

1. What is the wavefunction of the particle at time $t = 0$? [30]

2. What is the expectation value of momentum at time $t = 0$? [20]

3. What are the possible wavelengths of light emitted from the particle if the particle makes a transition to a lower energy state? [50]

Hint: The time-independent Schrodinger equation ($\widehat{H}\psi(x) = E\psi(x)$) for a particle with the potential energy:

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & elsewhere \end{cases}$$

has the following solution:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin(\frac{n\pi x}{L})$$

The quantum number n can be determined from the time-independent Schrodinger equation, if the energy is known.

$$\widehat{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)$$

$$\hat{p} = -i\,\hbar\frac{\partial}{\partial x}$$

$$h = 4.1\times10^{-15}\ eV\ s$$

$$c = 3.0\times10^{8}\ m/s$$